🤖 AI DATA ANALYST AGENT

Evidencia Técnica - Proyecto Portfolio ML Engineer

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📊 DESCRIPCIÓN

Herramienta web que analiza datasets CSV automáticamente:

- **Input: Subida CSV (Titanic 891x12)**
- **Output: Charts + métricas + preview en 5 segundos**
- **Demo: http://localhost:8501**

🎯 FEATURES IMPLEMENTADAS

✅ **Dashboard Metrics (filas/columnas/nulos)**

✅ **Histogram automático (1ra columna numérica)**

✅ **Preview DataFrame (head())**

✅ **JSON Dataset info (shape/memory/nulos)**

✅ **Responsive design (wide layout)**

✅ **Error handling (columnas numéricas)**

🛠️ TECH STACK

Frontend: Streamlit 1.38.0

Data: Pandas + NumPy

Charts: Plotly Express

Deployment: Local (streamlit run)

Entorno: Python 3.12 + venv

📈 RESULTADOS TITANIC DATASET

Shape: 891 filas × 12 columnas

Nulos: 866 (Age=177, Cabin=687)

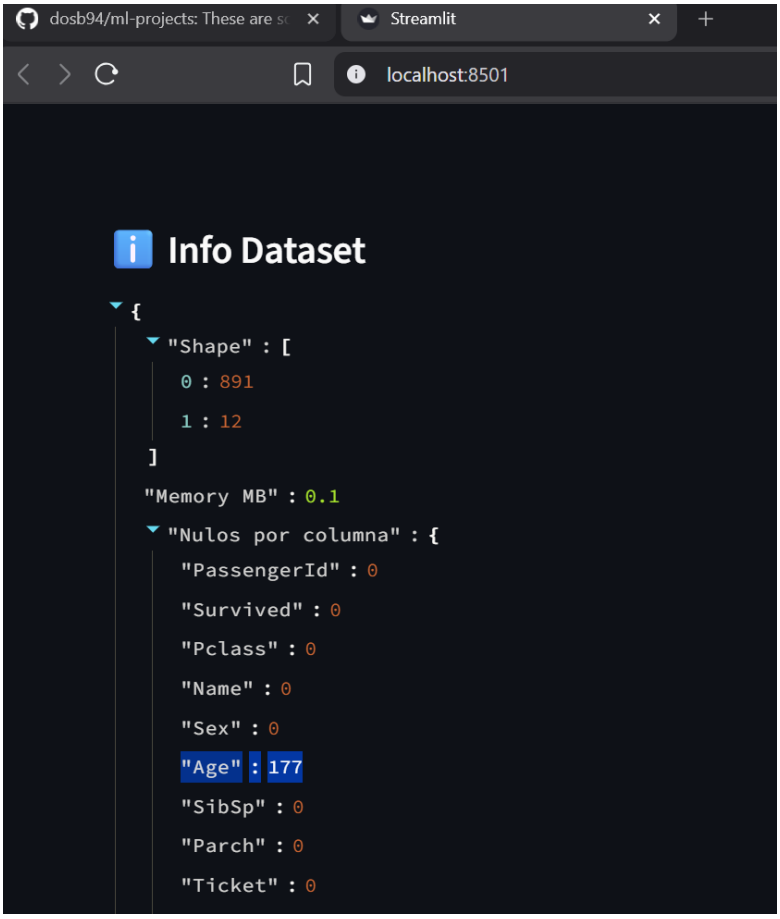
Memory: 85.5 KB

Columna analizada: Passenger Fare

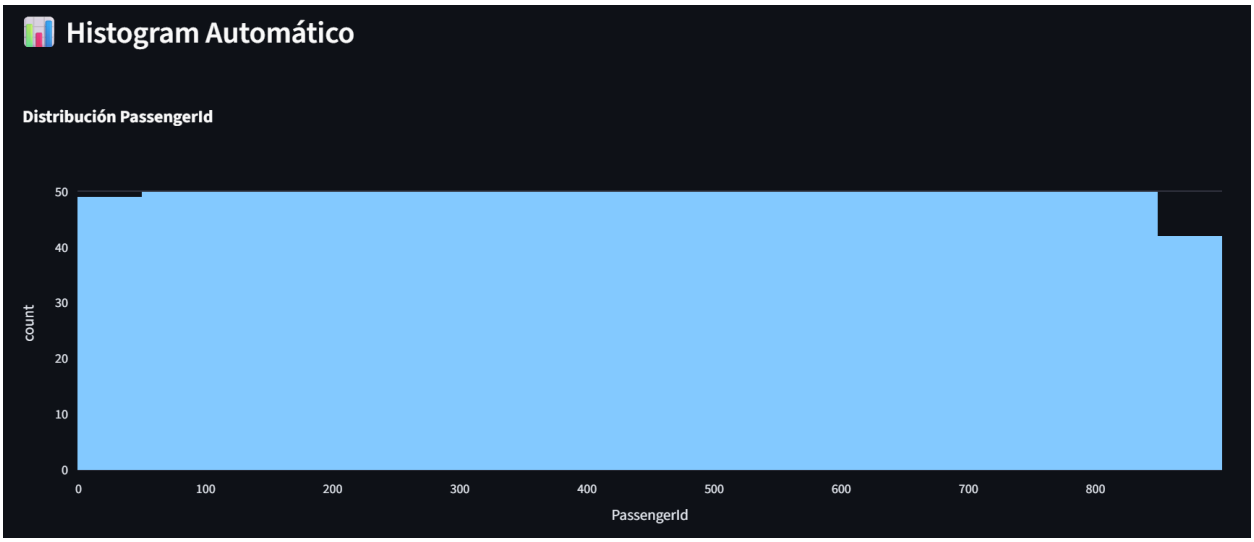
Screenshot Dashboard:

📋 Preview Dataset

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25	None	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3101282	7.925	None	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803	53.1	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35	0	0	373450	8.05	None	S



Screenshot Histogram:



💻 CÓDIGO CLAVE (app.py)

```
if uploaded_file:
        df = pd.read_csv(uploaded_file)
        st.success(f"✅ {df.shape}")
        # Metrics + Histogram + Preview
        fig = px.histogram(df, x=numeric_cols[0])
        st.plotly_chart(fig)
```

📞 CONTACTO

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